UNIVERSITY OF MESSINA

Bachelor's Degree in Computer Science L31 | Data Analysis

Tiered Edge AI Architecture for Cooperative UAV-Based Industrial Fire Detection

An onboard Sentinel-Scout system prototype

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Two-UAV platform and resource constraints

Heterogeneous onboard platforms with separate detection and verification roles

SENTINEL Customized X500 V2



Continuous detection role

D435i stereo hardware + Jetson Nano for onboard RGB inference

Larger X500 frame carries the sensing, compute and battery package

SCOUT ModalAI development platform



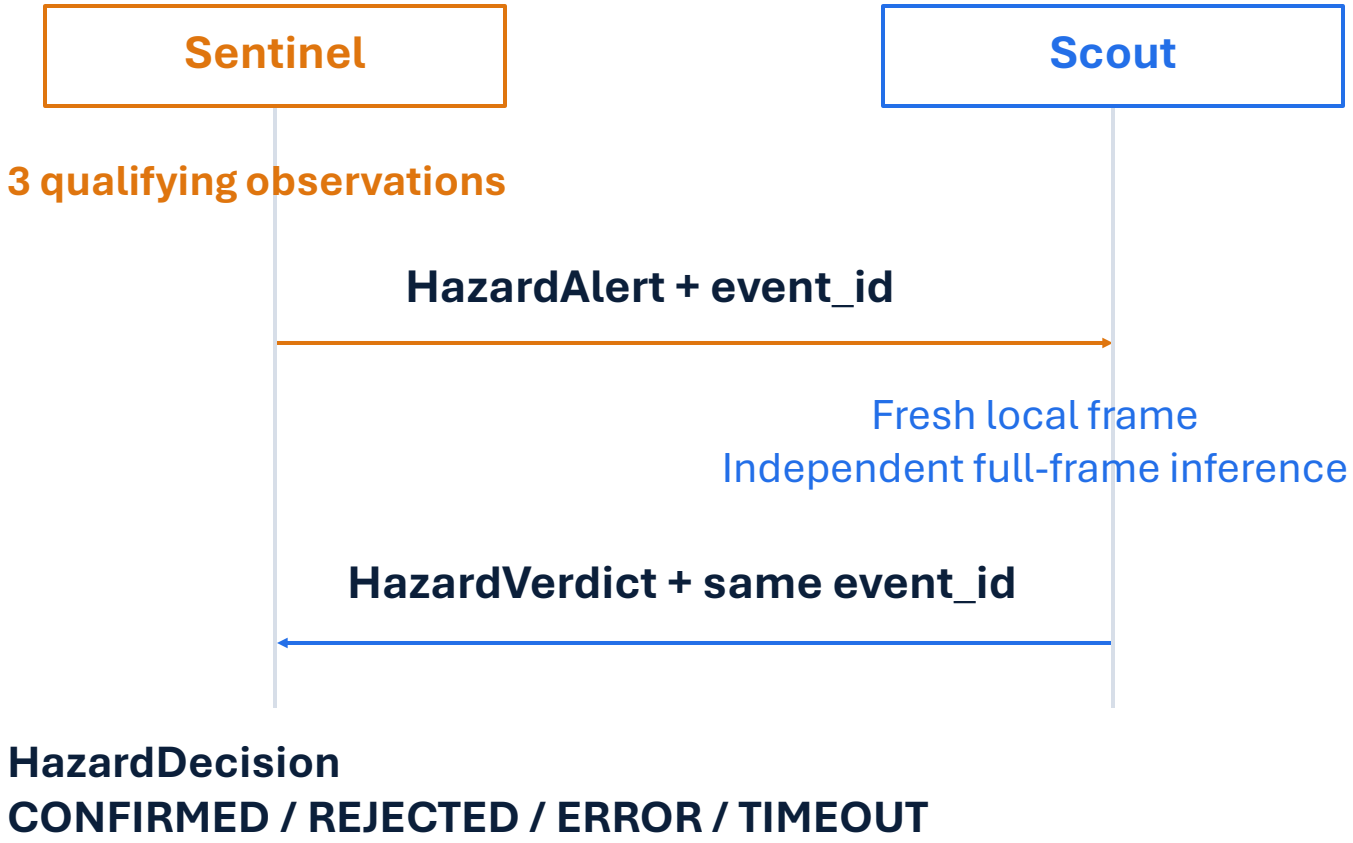
On-demand close verification

High-resolution Sony camera + VOXL 2 for independent inspection

Compact platform for intended targeted inspection

Correlated decisions and fault containment

Three-frame gating, independent verification and one terminal decision per event



10 / 10

Post-timeout verdicts rejected

0

Additional final decisions

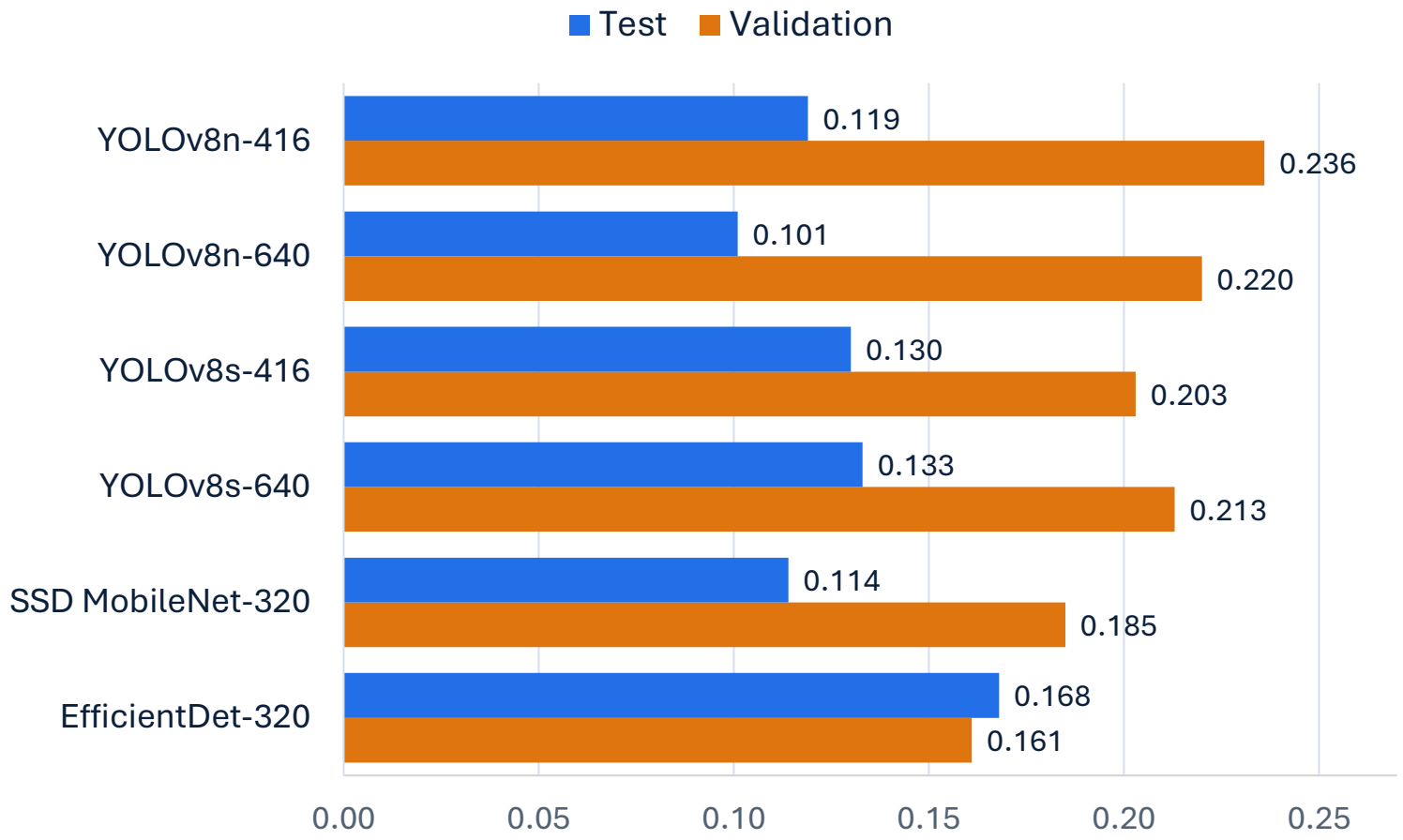
5 / 5

Unavailable-Scout events finalized as TIMEOUT

No ground-station inference. Image-space alerts carry no geographic coordinates.

Six-model evaluation on frozen data

AP50-95 on validation and the one-time held-out test set



3,558 / 700 / 733

Train / validation / test images

Grouped sources
Thresholds frozen before test

Best test AP: 0.168
EfficientDet-Lite0

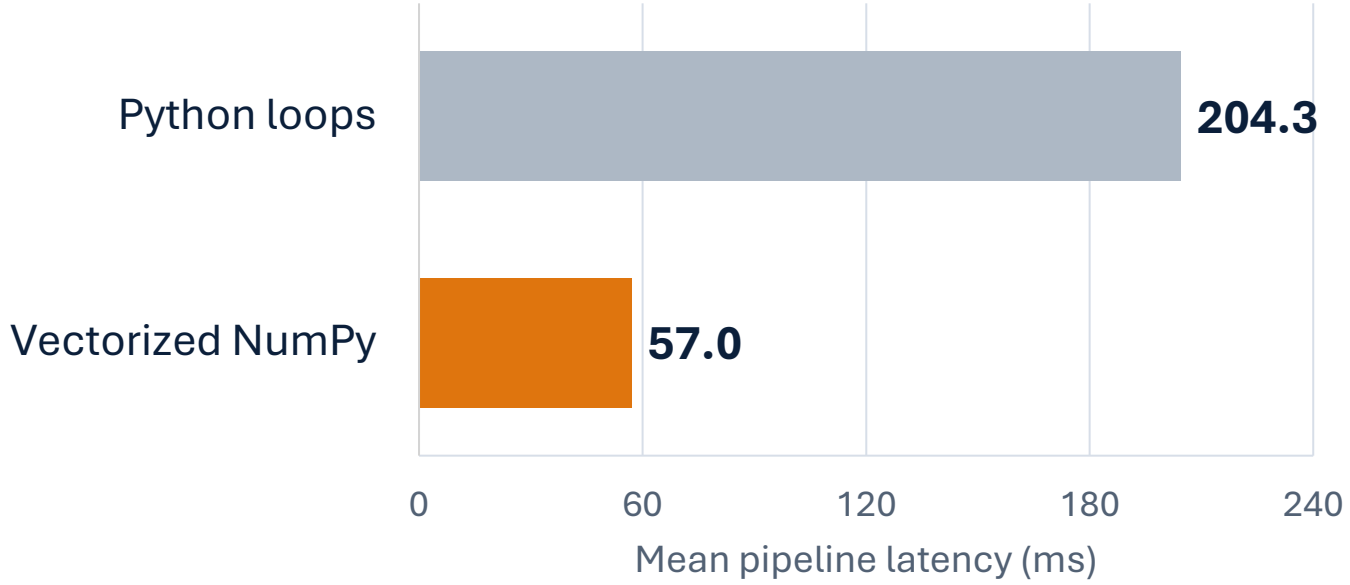
Six detectors evaluated with one fixed protocol and retained source records.

Predictive comparison informs the trade-off study. Earlier YOLO deployments use measured device implementations.

Vectorized decoding reduced pipeline time

Sentinel software experiment and separate device benchmarks

72.1% lower mean pipeline time



Sentinel decoding: 148.920 ms to 1.290 ms

8,400 candidates processed with NumPy vectorization

Separate July runs: 5 warm-ups, 50 measured pipelines each.

EXACT DEPLOYED PIPELINES

58.2 ms

Sentinel YOLOv8n
TensorRT FP16

p95 59.7 ms

999 ms

Scout YOLOv8s
TensorFlow Lite

p95 1.105 s

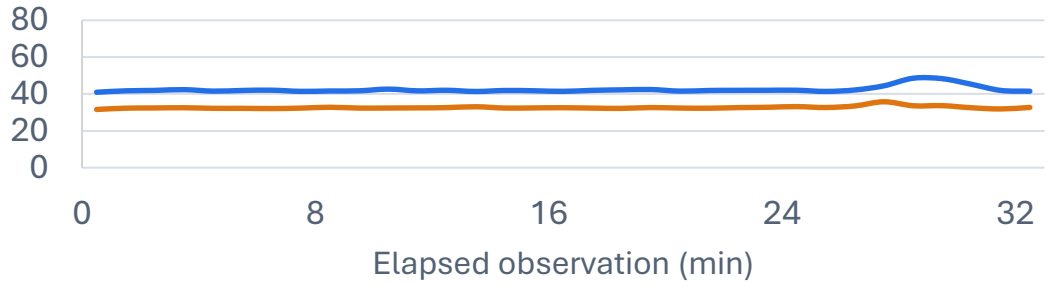
640 x 640 inputs, n = 100 per device
Camera and communication excluded

30-minute full-service resource evaluation

30,426 Sentinel frames. No recorded crash, OOM or thermal shutdown.

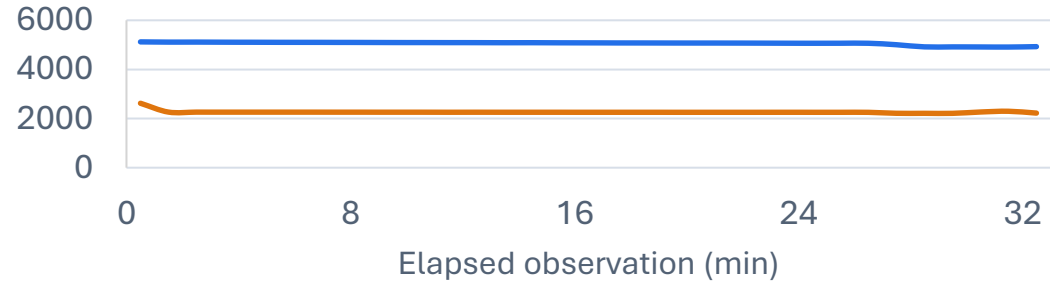
Sentinel Scout

Host CPU (%)



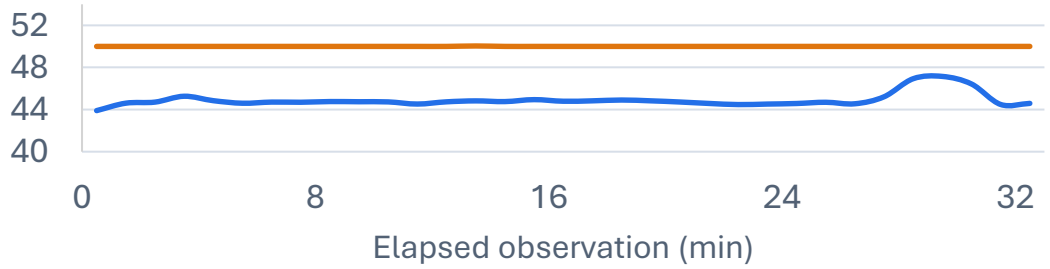
Raw maximum: 67.4 / 60.3%

Available memory (MiB)



Raw minimum: 1590 / 4874 MiB

Peak thermal-zone reading (°C)



Raw maximum: 51.0 / 48.2 °C

Swap used (MiB), raw trace



Raw change: +0.922 / 0.000 MiB

CPU, memory and temperature: 1-min means. Swap: raw. Extrema above use unaveraged samples.

Wireless detection and Scout verification

One stationary event linked across SwarmNet by the same event_id

SENTINEL / logged alert

0.174

Fire score at dispatch

3-frame temporal trigger

Final decision: CONFIRMED

Alert-to-decision: 1.471 s

Sentinel source image not retained

SCOUT / logged fire score 0.680



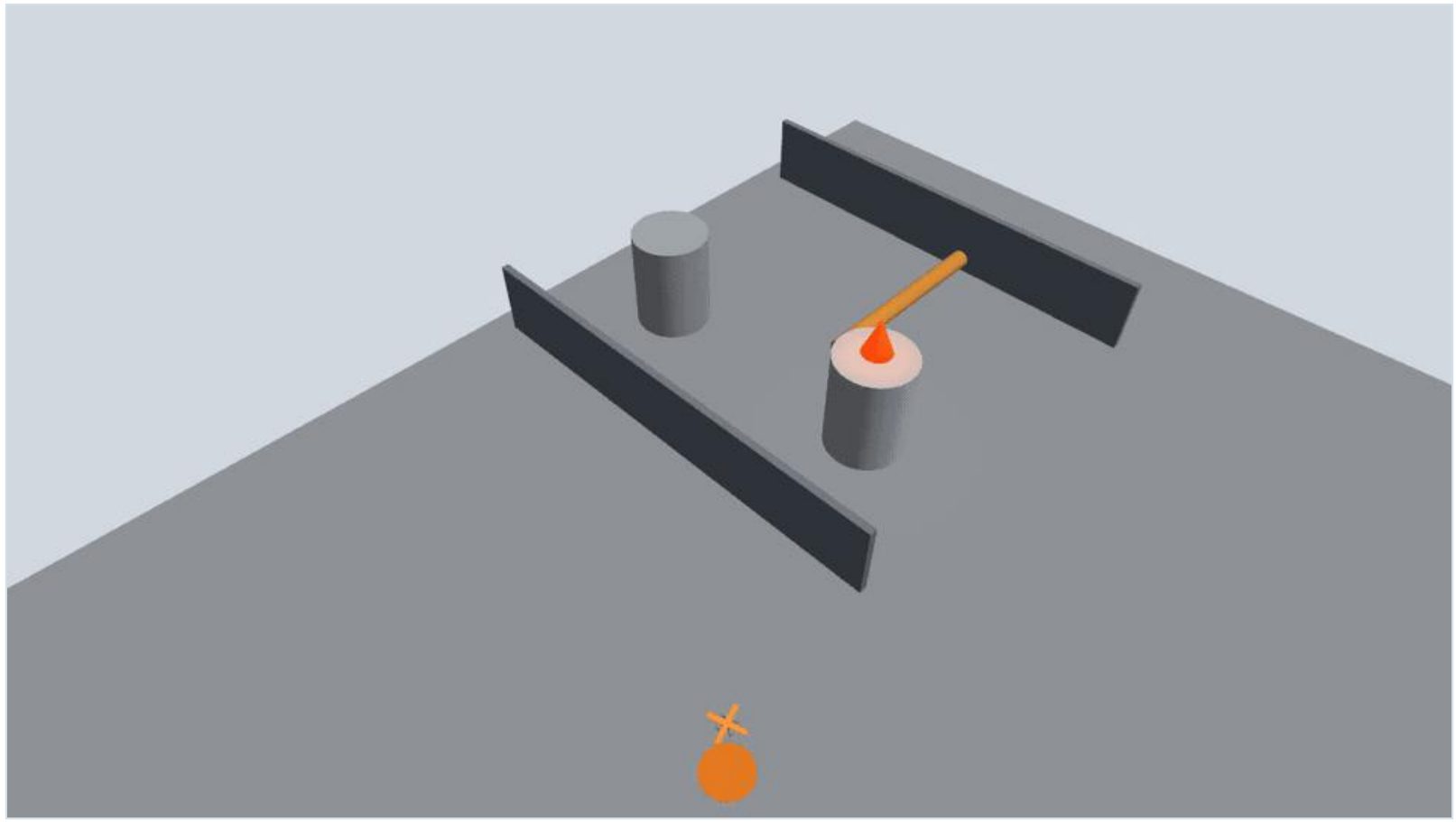
Loaded DDS: 1,000/1,000 replies, p95 RTT 7.984 ms

13/13 strong-flame confirmations. 0/6 direct unlit requests confirmed.

Flame box is an editorial highlight. Recorded model scores are not accuracy estimates.

Simulated aerial mission through ROS 2 and PX4

Sentinel patrol, Scout dispatch, verification and coordinated return



4.5 m

Commanded Scout standoff

214.253 s

Until both vehicles landed and disarmed in the log

Fault correction

0 Scout flight commands in corrected lost-alert and unavailable-Scout pilots

Scenario supplies trigger, target and verdict. Provisional airframes.

Engineering contributions and measured gains

A working onboard Sentinel-Scout system prototype

1 Heterogeneous integration

Independent camera inference on both UAV computers
Containerized and native ROS 2 Foxy on SwarmNet

2 Traceable, fault-contained decisions

10/10 late verdicts rejected, no extra final decision
5/5 no-frame and 5/5 stale-frame error verdicts

3 Measured computational improvement

72.1% lower mean Sentinel pipeline time
>30 min concurrent-service observation on both devices

4 Executable mission orchestration

ROS 2/PX4 simulated mission with a 4.5 m standoff
Both vehicles landed and disarmed in the retained log

Limits: small flame 1/10. Scout p95 2.618 s > 2.0 s. Jetson swap +0.922 MiB.

Next: controlled accuracy ablation, external evaluation and camera-driven aerial validation.



Questions

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Mohammadamin Jahanimajd

University of Messina | Academic Year 2025/2026